DAY: ________ DATE: ________

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Topic: Types of Frequency distribution:

Class Limits: Class limits is defined as:
"The end points / values / Scores of a class called class limits."

Types:

Lower class limits (LCL) → starting value of the class.

Upper class limits (UCL) → Ending value of the class.

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Example Table:

Class interval	Class limits	Frequency (F)
86–90	86, 90	6
91–95	91, 95	4
96–100	96, 100	10
101–105	101, 105	6
106–110	106, 110	3
111–115	111, 115	1
Total:		30

Class Boundaries: Class boundaries are defined as:

"Class boundaries which are obtained to remove the gap between consecutive classes"

Formulas: Lower class Boundary (LCB) = LCL − (Gap ÷ 2)

Upper class Boundary (UCB) = UCL + (Gap ÷ 2)

Example:

For 86–90 → Boundaries = 85.5 — 90.5

Lower class Boundary = $l_1 - \frac{d}{2}$

Upper class Boundary = $l_2 + \frac{d}{2}$.

For 91—95 → Boundaries = 90.5—95.5.

Class Boundaries Table:

Class Interval	Class Boundaries
86—90	85.5—90.5
91—95	90.5—95.5
96—100	95.5—100.5
101—105	1005—105.5
106—110	105.5—110.5
111—115	110.5—115.5.

Class Marks: The class marks is the middle value of a class.

Formula:

$$X = \frac{LCL + UCL}{2}$$

Example:

For 86—90 = $\frac{86+90}{2}$ = 88

For 91—95 = $\frac{91+95}{2}$ = 93.

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Class Intervals: This is defined as: "The gap between two consecutive classes is called the class intervals".

Formula:

Class interval = UCL − LCL = Difference b/w

Example: For 86 − 90 →
interval = 90 − 86 = 04 (or 90.5 − 85.5) = 5 depending on class boundary method.

Key Notes:

Frequency: Number of values falling in a particular class.

Class Boundary: Ensure continuity in data.

Class marks: Used for further statistical calculation (Mean, Varience etc)

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Class marks	Class limits	Frequency (F)	Class Boundaries
$\frac{86+90}{2} = 88$	86 — 90	6	85.5 — 90.6
$\frac{91+95}{2} = 93$	91 — 95	4	90.5 — 95.5
$\frac{96+100}{2} = 98$	96 — 100	10	95.5 — 100.5
$\frac{101+105}{2} = 103$	101 — 105	6	100.5 — 105.5
$\frac{106+110}{2} = 108$	106 — 110	3	105.5 — 110.5
$\frac{111+115}{2} = 113$	111 — 115	1	110.5 — 115.5.
	Total	30	